AMEYA NITIN PHADNIS

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EDUCATION

Worcester Polytechnic Institute (WPI), Worcester, MA — May 2026
Master of Science (M.S.) in Robotics Engineering

Manipal Institute of Technology, Manipal, India — August 2023
Bachelor of Technology (B.Tech) in Electrical and Electronics Engineering

WORK EXPERIENCE

Research Assistant, Adaptive and Intelligent Robotics (AIR) Lab, WPI, *Worcester, MA* — Feb 2026 - Sept 2026
***Skills**: Python, C++, MoveIt2, Open3D, ROS2, 3D Perception, Depth Cameras, Point Cloud Processing, Docker, MuJoCo, Robotic Manipulation*

- Built an **adaptive Octree-based grasp-planning** pipeline that processed point cloud from **RGB-D camera**. Developed an **adaptive Octree partitioning** algorithm to **identify grasp regions** on the objects and generate grasp poses using the **antipodal grasp condition**. Work submitted to **ICRA 2027**.
- **Integrated** the algorithm with **Panda Robotic Arm** and **Intel RealSense Depth Camera**, with testing in **MuJoCo sim** with **ROS 2**, **MoveIt2** integrations.
- **Benchmarked** the developed algorithm against **GPD** and **GraspGen** on **GPU compute,** demonstrating a **30%** more efficient and **robust grasp-planning**.

Perception Co-op, Symbotic LLC, *Wilmington, MA* — Sept 2025 - Dec 2025
***Skills**: Python, ROS2, Open3D, 3D Perception, LiDAR, Depth Camera, Sensor Calibration, Hardware Integration, Docker, Mobile Robots*

- Built a robust and efficient **Python module** for performing **online auto-calibration** with **purely 3D data** from **LiDAR**, **depth cameras** for mobile robots in **warehouse environments**. **Testing** and **integration** on hardware with Docker and **ROS2** was done achieving sensor orientation **accuracy** within **0.5 degree**
- Designed the calibration algorithm by using **point cloud processing pipelines** like plane fitting, **noise reduction** and feature extraction for accurate results.
- Deployed **data collection scripts** and **tested software** on **NVIDIA Jetson TX2** robot hardware with **ROS 2**, **Docker**, and **LiDAR integration.**
- Developed robust **tools** in **Python** for **performance validation** of **Collision Avoidance** systems deployed on production. This involved **extracting essential data** from multiple modules using **data scripts** integrated with **DVC, GCP** and **Snowflake** and building effective **visualizations** for reliable **data analysis**.

Computer Vision Intern, DEKA Research and Development Corp., *Manchester, NH* — May 2025 - Aug 2025
***Skills**: OpenCV, Computer Vision, Python, Image Processing, Infrared Cameras, Medical Devices, Raspberry Pi, Custom Dataset Generation*

- Designed an efficient **vision system** for drop volume estimation using an **infrared camera** for a medical IV infusion device on limited compute.
- Implemented an **optimized Sobel Edge Detection** and **Hough Transform** algorithms to **improve drop volume** estimation by **10%**.
- Developed an algorithm to **auto-annotate** drop images to train a **Spline Estimation Machine Learning** model for precise detection in **harsh conditions**.

Graduate Researcher, Adaptive and Intelligent Robotics (AIR) Lab, WPI, *Worcester, MA* — Jan 2025 - May 2025
***Skills**: Python, OpenCV, Open3D, 3D Perception, Depth Cameras, Machine Learning, Point Cloud Processing*

- Designed a **novel vision** pipeline that performs **optimized and accurate 3D mapping** of indoor objects using **RGB Depth Camera** for **accurate grasping**.
- Implemented the **isolation-Forest** algorithm for filtering and **optimized Octree spatial partitioning** using compact **voxels** to increase **point cloud** details.
- **Benchmarked** the pipeline on the **YCB Dataset**, achieving a **25%** more **optimized** Octree-partitioned point cloud without loss of object details.

Software Engineer, Bajaj Finance Ltd., Bajaj Finserv, *Pune, India* — July 2023 - July 2024
***Skills**: Java, REST APIs, Middleware Design, Microsoft Azure, Microservices, Kafka, CI/CD, Docker, Unit Testing, Object-Oriented Design*

- Developed and deployed **production-quality software** for the Personal Loans module serving more than **4 million customers**. Implemented new **REST API** calls, **database connections**, and business logic for a major user-experience revamp that increased **revenue by 7%**.
- Designed, developed, and deployed a production microservice that synchronized with a Chatbot and used a **multi-threaded pipeline** to upload user information to remote **Azure Blob** Storage, enabling the Chatbot launch and increasing customer **reach by 5%**.
- Developed **robust unit tests** using JUnit framework for production microservices, and integrated them with **SonarQube** and into the **CI/CD** pipeline

PROJECTS

Motion Planning in Dynamic Environments Using Dynamic Roadmaps (GitHub)
***Skills:** Open Motion Planning Library (OMPL), Python, C++, PyBullet Simulation, Boost, Sampling-based planning, Robotic Manipulation*

- **Optimized motion planning** in a dynamic environment by designing a **Dynamic Roadmaps** algorithm to perform **table-pick** on various **robotic arms** in **PyBullet Simulation**. The planner **outperformed** the PRM planner in **planning time** for **dynamic obstacles**, with a **70% reduction** in planning time.

Autonomy Stack for an Autonomous Formula SAE Vehicle (Project Page)
***Skills:** Python, OpenCV, ROS, Sensor Fusion, Perception, Object Detection, Localization, SLAM, Gazebo Simulation, Autonomous Vehicles*

- Led the **design and implementation** of a Formula SAE **autonomous vehicle**. Built the **Perception** module using **RGB Cameras** using **YOLO Object Detection** and PnP-based depth estimation and an **EKF-based Localization** and **SLAM** module using **sensors** like **GNSS**, **IMU**, and **Wheel Odometry**.

Perception and Environment Reconstruction for Autonomous Vehicles using Monocular Camera (Project Report)
***Skills:** Python, OpenCV, Computer Vision, PyTorch, Deep Learning, Blender Simulation, Scene Understanding*

- Developed a vehicular **perception pipeline** by analyzing, tuning and applying **state-of-the-art deep learning algorithms** on images from a RGB camera.
- Integrated algorithms to perform **object detection** using **YOLO** and **DETiC**, **depth estimation** using **MiDaS**, **semantic segmentation** using **Mask-RCNN**, **pose estimation** of surrounding vehicles and pedestrians using **Faster R-CNN**. A **Blender** simulation/**visualization** displayed all surrounding information.

Visual Inertial Odometry using MSCKF and Deep Learning (Project Report)
***Skills:** Python, OpenCV, PyTorch, Computer Vision, Sensor Fusion, Localization, Deep Learning, State Estimation, Robot Navigation, Synthetic Dataset Generation*

- Developed an efficient **Python** implementation for performing **drone localization** using the **Multi-State Constraint Kalman Filter** through **sensor fusion** of **stereo camera** and **IMU** data from the EuRoC MAV dataset. Implemented **essential filter functionalities**, achieving **RMSE of 1.5** in drone pose.
- Built a **robust dataset generation pipeline** and **trained** multiple **deep learning** architectures (CNN, RNN, Transformer based) for visual-inertial odometry

Camera-based 3D Scene Reconstruction using Structure-from-Motion (SfM) (GitHub)
***Skills:** Python, OpenCV, 3D Computer Vision, 3D Geometry, 3D Reconstruction, Feature Extraction, Epipolar Geometry, Advanced Image Processing*

- Developed a Python-based implementation of the **SfM algorithm** pipeline for scene reconstruction and **camera pose estimation** for **monocular camera images**. The pipeline's main features involved performing **RANSAC**, **Triangulation**, **Perspective-n-Point**, and **Bundle Adjustment** for accurate scenes.

3D Scene Reconstruction using Neural Radiance Fields (NeRF) (GitHub)
***Skills:** PyTorch, Python, Deep Learning, 3D Computer Vision, Neural Rendering, 3D Reconstruction using Images*

- Designed an end-to-end **PyTorch** implementation of the **NeRF algorithm** for performing **3D reconstruction**. The code leveraged **deep neural networks, volume rendering** and **hierarchical volume sampling**, and successfully performed **reconstruction of 2 objects**, with an average **PSNR** of 27.75 and 22.32.

Homography Estimation using Deep Learning for Image Stitching (GitHub)

Skills: *Python, OpenCV, PyTorch, Computer Vision, Deep Learning, Image Processing, Custom Dataset Creation*

- Designed and trained **PyTorch**-based **homography estimation** and **image stitching** deep **neural networks** with a **robust dataset generation pipeline**. Two **CNN** models were trained using supervised and unsupervised learning approaches on a dataset created from MSCOCO for successful image stitching.

Autonomous Navigation of TurtleBot3 using Deep Reinforcement Learning (GitHub) Nov 2024 - Dec 2024

Skills: *Python, PyTorch, ROS2, Deep Learning, Reinforcement Learning, Gazebo Simulation, Autonomous Navigation, Mobile Robots*

- Built an end-to-end pipeline for the **TD3 Deep Reinforcement Learning Algorithm** to perform autonomous navigation on the **TurtleBot3 mobile robot** in a **custom Gazebo simulation** with obstacles using **ROS2** and **PyTorch**. The agent was **successfully** able to navigate with an **accuracy** of **75%.**

TECHNICAL SKILLS

- ***Programming Languages*:** Python, C++, Java, MATLAB, Simulink
- ***Software Development***: ROS/ROS2, Linux, Git, Docker, Google Cloud Platform (GCP), DVC, Microsoft Azure
- ***Relevant Libraries/Frameworks****:* PyTorch, Tensorflow, MoveIt2, OMPL, OpenCV, Open3D, Gazebo, MuJoCo, PyBullet